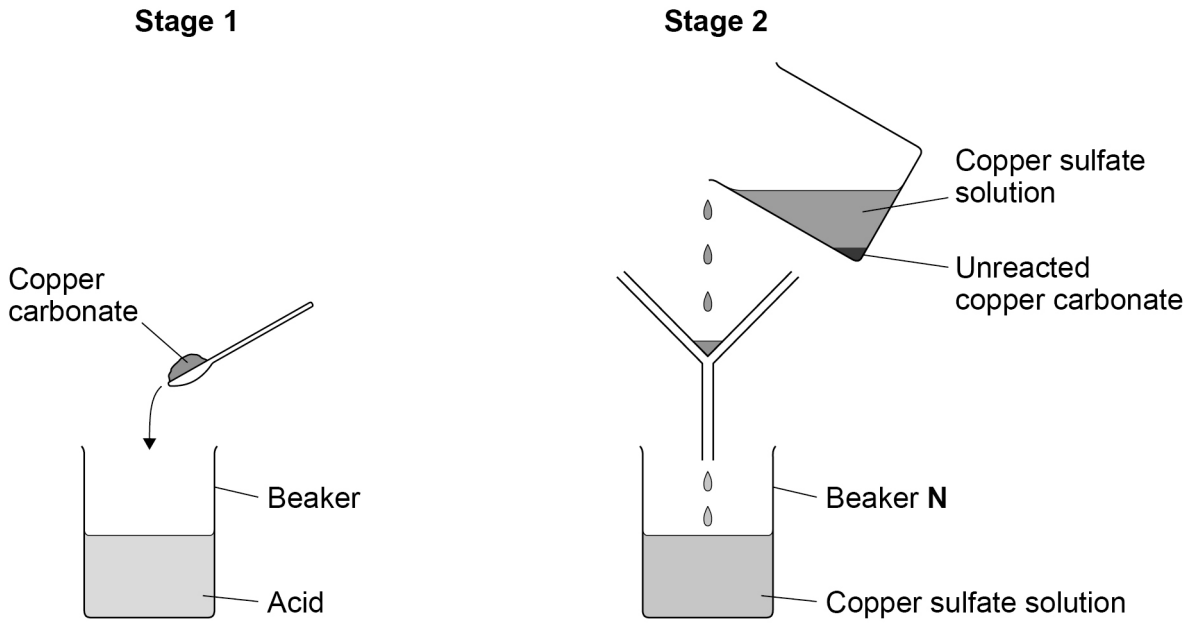
Do not write outside the box

0 2 A student prepared copper sulfate by reacting an acid with excess copper carbonate.

Figure 3 shows the first two stages in the preparation of copper sulfate.

Figure 3



0 2 . 1 What is the formula of the acid used to prepare copper sulfate?

[1 mark]

Tick (✓) **one** box.

HCl ☐

HNO_3 ☐

H_2SO_4 ☐

Do not write outside the box

0 2 . 2 Why is excess copper carbonate used in **stage 1**?

[1 mark]

0 2 . 3 Beaker **N** contained copper sulfate solution.

Describe how the student could produce copper sulfate crystals from the copper sulfate solution in beaker **N**.

[2 marks]

Question 2 continues on the next page

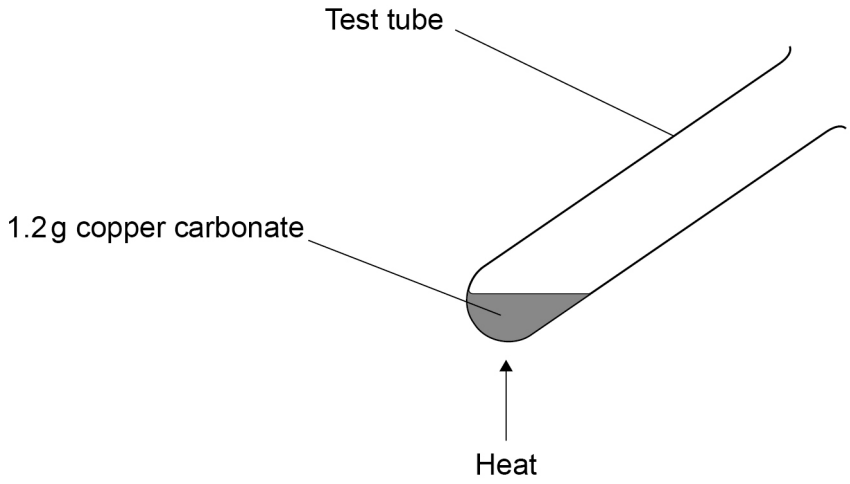
Do not write outside the box

A student investigated the thermal decomposition of copper carbonate.

Copper carbonate decomposes to form two products.

Figure 4 shows the apparatus.

Figure 4



This is the method used.

1. Add 1.2 g of copper carbonate to a test tube.
2. Heat the test tube and contents until the mass does not change.
3. Record the mass of the contents of the test tube after heating.
4. Repeat steps 1 to 3 with different masses of copper carbonate.

Table 1 shows the results.

Table 1

Mass of copper carbonate in test tube before heating in grams	Mass of the contents of test tube after heating in grams
1.2	0.8
2.4	1.7
3.6	2.2
4.8	3.1
6.0	3.9

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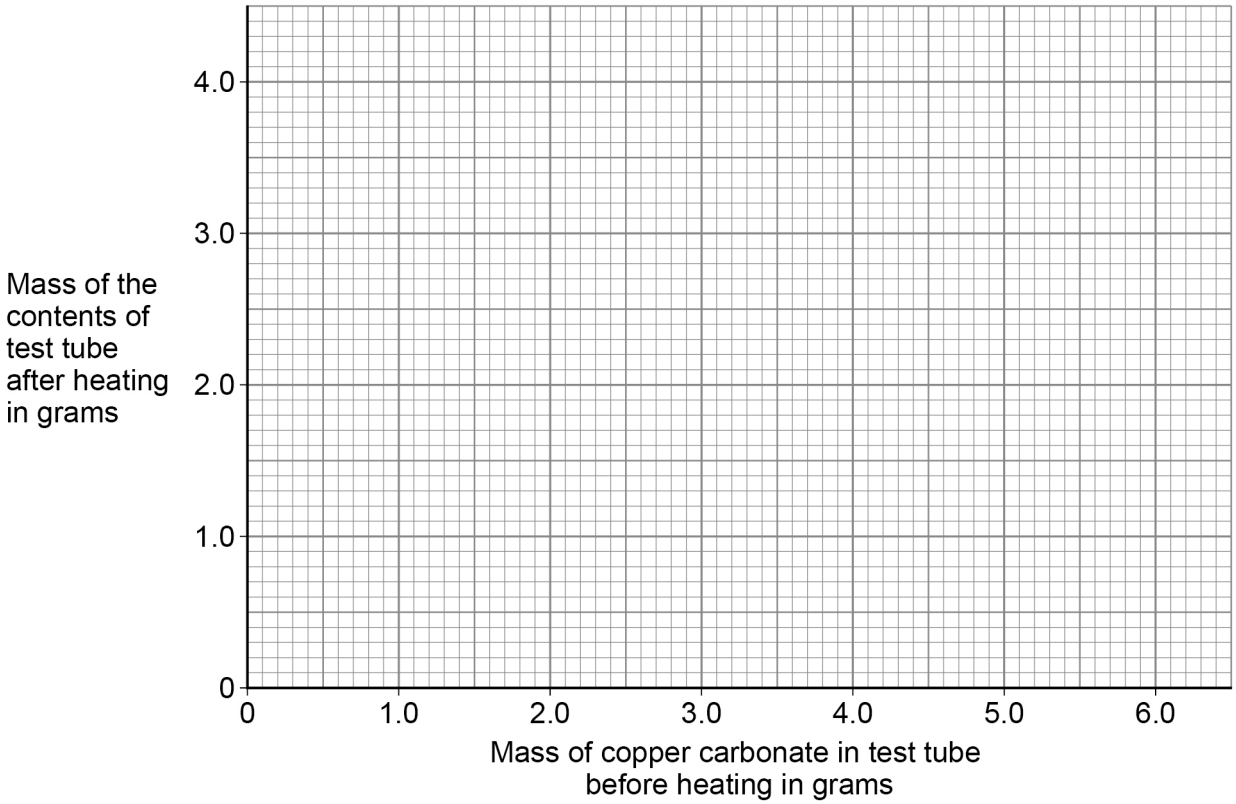
0 2 . 4 Plot the data from **Table 1** on **Figure 5**.

Draw a line of best fit.

[3 marks]

Figure 5



0 2 . 5 Why does the mass of the contents of the test tube decrease in mass when copper carbonate is thermally decomposed?

[1 mark]

_____

_____

8

Turn over for the next question